Online Supplement

Source-Learned Structural Priors for Chance-Constrained Simulation Optimization

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EC.1. Frozen Protocol and Accounting

Table EC.1 The primary source-to-target information contract.

Source tasks	8
Public candidates per source task	64
Source repetitions per candidate	12
Target initial points	8
Target search points	24
Independent verification repetitions	928
Mean/risk heads	f, g, $\eta = \log \mathrm{Var}(Y^g)$
Dictionary size	68 centered/scaled functions
Original confirmation interval	not run as a block; seed 95434 overlaps development

Source calls are 8×64×12= 6,144. The paired source-structure control uses two model arms but the same keyed target stream; its additional local execution made 4,608 simulator calls for the two arms and eight tasks. These are paired development calls, not independent confirmation observations. Oracle moments are read only after recommendation for scoring and are never supplied to fitting or acquisition.

EC.1.1. Public coordinates and the finite dictionary

Each candidate is an ordered inventory policy $p = (p_1, p_2, p_3, p_4) \in [0,1]^4$. With capacity $C = 20$ and initial stock $s_0 = 8$, the observed stock bin is $b_t = \min\{\lfloor 4s_t/C \rfloor, 3\}$, the order is $o_t = \max\{Cp_{b_t+1} - s_t, 0\}$, and the next stock is $s_{t+1} = \max\{s_t + o_t - d_t, 0\}$. For the declared twelve-period demand schedule, the local trajectory weights are

$$w_{t1} = o_t/(CT), \qquad w_{t2} = s_{t+1}/(CT), \qquad w_{t3} = \max\{d_t - s_t - o_t, 0\}/\sum_t d_t.$$

Let $A_j = \|w_{\cdot j}\|_2$ for $j = 1,2,3$ and $(N_1, N_2) = (\sum_t w_{t1},\ \sum_t w_{t3})$ for the two shared volume exposures used by the simulator. The public coordinate is

$$\psi(p) = (A_1, A_2, A_3, N_1, N_2, h, \nu, s_T/C),$$

where $h = \sum_t w_{t2}$, $\nu = T^{-1}\sum_t \mathbb{1}\{o_t > 0\}$, and s_T is terminal stock. Thus the eight coordinates are observable from the trajectory and do not contain prices, outcomes, or a hidden optimum.

The implementation uses a 68-function dictionary. After pooled source standardization, it contains the eight coordinates ψ_j, their eight squares, the sixteen $\sin(\pi\psi_j)$ and $\cos(\pi\psi_j)$ terms, the 28 pairwise products $\psi_i\psi_j$ $(i<j)$, and the eight high-frequency controls $\sin(4\pi\tanh(\psi_j/2))$. The source learner scores these fixed functions by low-frequency smoothness, cross-task relevance, sign consistency, and a pilot cross-validation loss, then greedily retains four functions per response head. The full-dictionary arm retains all 68 functions; the target-only arm uses the same 68 functions with a zero-mean coefficient prior and no source labels. No trajectory encoder or learned input coordinate is used.

EC.1.2. Source prior, target update, and verifier

For each mean response head $t \in \{f, g\}$, source labels are fitted by weighted ridge regression on the selected standardized functions. The source task coefficient vectors supply a head-specific mean and covariance; the target coefficient mean starts at zero in the target-only control. Paid target labels update the Gaussian linear model with the observed diagonal noise. The physical-risk head additionally updates a finite factor-family particle mixture by the exact Gaussian log-sample-variance likelihood. The risk head is the total log variance $\eta = \log \mathrm{Var}(Y^g)$; its simulator law is

$$\mathrm{Var}(Y^g) = \rho_0 + A^\top \Lambda A + N^\top B N + N^\top \omega.$$

The observation at a policy uses 12 independent episodes. Its objective and constraint means have noise variances $\sigma_f^2/12$ and $\widehat{v}_g/12$; the log-sample-variance label uses the digamma bias correction and noise $\psi_1((12-1)/2)$.

The acquisition rule uses a three-dimensional Gaussian approximation for (f, g, η) and a 3-node Hermite rule per head, hence 27 joint fantasy nodes. For a candidate x, the reported deployment utility is

$$U(x) = \Pr\{g(x) + z_{.95}\exp(\eta(x)/2) \le 0\}\,\mathbb{E}\left[\frac{5 - f(x)}{10}\right],$$

with objective quality clipped to the declared $[-5, 5]$ simulator range. The 928-repetition verifier is independent of paid observations and returns zero realized utility when its chance certificate fails. These definitions make the target-only baseline and the source-selected arm identical in candidate bank, budget, noise streams, acquisition, and verification.

EC.2. Source-Structure Control

Table EC.2 gives the complete paired final-reward comparison for the eight development tasks. Both arms use observable inventory exposure, the same pooled factor-family risk prior, the same initial points, the same Gaussian KG rule, and the same verifier.

The positive-difference count is 3/8, the negative count is 1/8, and four tasks are tied. The development contract therefore records an attribution signal but fails the preregistered stable-deployment gate.

Table EC.2 Taskwise source-selected structure minus same-source full dictionary.

Law seed	Initial reward	Structure	Full dictionary	Difference	Both certified
95425	0.875956	0.881686	0.881686	+0.000000	yes
95426	0.880274	0.880274	0.878995	+0.001279	yes
95427	0.785842	0.859288	0.859288	+0.000000	yes
95428	0.855284	0.871915	0.873167	-0.001253	yes
95430	0.824105	0.851198	0.851198	+0.000000	yes
95431	0.795239	0.795239	0.751917	+0.043322	yes
95432	0.848587	0.848587	0.848340	+0.000247	yes
95434	0.898958	0.901946	0.901946	+0.000000	yes
Mean		0.861267	0.855817	+0.005449	8/8

EC.3. Risk and Representation Diagnostics

The observable coordinate reduces eight-point holdout objective-mean RMSE from 0.594 to 0.270 and constraint-mean RMSE from 0.204 to 0.141. Constraint-rank Spearman correlation increases from 0.119 to 0.624, while the initial true recommendation is feasible in 7/8 tasks for both coordinate choices.

The factor-family risk prior reduces initial target η RMSE from 0.363 to 0.301 on all eight development tasks. Its coefficient RMSE is 0.255 versus 0.251 for pooled risk, and its closed-loop mean final reward is 0.8599 versus 0.8605. These numbers separate total-risk prediction from factor recovery and deployment value.

EC.4. Same-Contract Target-Only Baseline

The target-only full-dictionary arm uses the same 68-function basis, target initial design, paid-observation budget, keyed noise streams, pooled risk arm, acquisition rule, and independent verifier. Its only difference is that it starts without source labels. The complete comparison is:

Table EC.3 Same-contract target-only baseline. Parentheses give certified tasks.

Block	Source-selected mean (certified)	Full dictionary mean (certified)	Target-only mean (certified)	RBF-KG mean (certified)	Source–RBF mean difference
Development	0.86127 (8/8)	0.85582 (8/8)	0.76199 (7/8)	0.86239 (8/8)	-0.00112
Independent v2	0.44274 (5/8)	0.45160 (4/8)	0.45148 (4/8)	0.45912 (4/8)	-0.01638

On the development block, source-selected structure, same-source full dictionary, and target-only full dictionary obtain mean final rewards 0.86127, 0.85582, and 0.76199, respectively. On the independent v2 block the three means are 0.44274, 0.45160, and 0.45148. The target-only arm therefore provides the required feasible nonempty comparator, while the independent block still does not support a stable source-structure deployment advantage.

For a generic constrained-BO control, we additionally froze an isotropic RBF kernel (lengthscale 0.25, outputscale 1.0) and fit all three heads only to target observations. This target-only RBF-KG arm obtains

means 0.86239 and 0.45912 on development and v2, respectively. It is therefore a stronger generic control for decision ranking than the source-selected arm on both blocks; the source-structure attribution remains a comparison with the same-source full dictionary, not a claim of generic BO superiority.

EC.5. Paired Uncertainty Summaries

All taskwise differences reported in the manuscript are accompanied by a seed-level paired percentile bootstrap with 100,000 resamples. These intervals are descriptive because each block has eight law seeds; they are not a test of universal superiority.

Table EC.4 Paired bootstrap intervals for reported differences.

Metric	n	Mean difference	Percentile bootstrap 95% interval
observable minus policy objective RMSE	8	-0.32418	[-0.48330, -0.16914]
observable minus policy constraint RMSE	8	-0.06390	[-0.12552, -0.00427]
observable minus policy log-risk RMSE	8	+0.00000	[+0.00000, +0.00000]
observable minus policy constraint rank correlation	8	+0.50491	[+0.19041, +0.77954]
learned minus pooled log-risk RMSE	8	-0.06202	[-0.08830, -0.04120]
learned minus pooled factor coefficient RMSE	8	+0.00439	[-0.03358, +0.03654]
development: source_selected_structure_minus_same_source_full_dictionary	8	+0.00545	[-0.00031, +0.01641]
development: source_selected_structure_minus_target_only_full_dictionary	8	+0.09928	[-0.00044, +0.29825]
development: same_source_full_dictionary_minus_target_only_full_dictionary	8	+0.09383	[-0.00048, +0.28197]
development: source_selected_structure_minus_target_only_rbf_kg	8	-0.00112	[-0.00476, +0.00129]
development: same_source_full_dictionary_minus_target_only_rbf_kg	8	-0.00657	[-0.02106, +0.00126]
development: target_only_full_dictionary_minus_target_only_rbf_kg	8	-0.10040	[-0.30293, +0.00136]
independent_v2: source_selected_structure_minus_same_source_full_dictionary	8	-0.00886	[-0.02879, +0.00184]
independent_v2: source_selected_structure_minus_target_only_full_dictionary	8	-0.00873	[-0.02879, +0.00197]
independent_v2: same_source_full_dictionary_minus_target_only_full_dictionary	8	+0.00012	[-0.00002, +0.00039]
independent_v2: source_selected_structure_minus_target_only_rbf_kg	8	-0.01638	[-0.04255, +0.00738]
independent_v2: same_source_full_dictionary_minus_target_only_rbf_kg	8	-0.00752	[-0.02714, +0.00939]
independent_v2: target_only_full_dictionary_minus_target_only_rbf_kg	8	-0.00765	[-0.02726, +0.00933]

EC.6. Claim Audit

Table EC.5 Permitted interpretation of each main claim.

Claim	Evidence status	Manuscript wording
Source function structure improves early prediction	supported	Claim within the declared representation and dictionary.
Target observations correct source mismatch	supported	State that component weights can be downweighted.
Risk structure is learned	partial	Report total-risk calibration only.
Risk factors are separately recovered	unsupported	Keep outside contributions.
Stable independent deployment advantage	unsupported	Report the failed 3/8 gate and negative controls.
General high-dimensional free-variable optimization	unsupported	Describe ordered profiles and nominal dimension only.
Zero false certificates are universal safety	unsupported	Restrict the statement to the declared verifier law.

Table EC.6 Independent fixed-seed confirmation block; all eight tasks retained.

Arm	Tasks	Mean final reward	Certified	False certificates
Source-selected structure	8	0.44274	5/8	0
Same-source full dictionary	8	0.45160	4/8	0

EC.7. Reproducibility Pointers

The source archive and development artifact are in `paper_artifacts/meta_prior_feasible_deployment_v1/`. The paired control is regenerated by `performance/run_meta_prior_feasible_source_arm_v1.py` from `performance/meta_prior_feasible_source_arm_v1.json`. The same-contract target-only controls are regenerated from `performance/meta_prior_feasible_baseline_development_v1.json` and `performance/meta_prior_feasible_baseline_v1.json`. The baseline table and paired bootstrap interval table are generated by `performance/render_meta_prior_baseline.py` and `performance/summarize_meta_prior_uncertainty.py`; neither script starts a simulator. The original confirmation interval was not run as a block. Seed 95434 overlaps the development artifact and is excluded from any clean-confirmation claim. The separate v2 block is regenerated from its own frozen contract and retains all eight new seeds; it is reported as an independent bounded check, not as a replacement for the original block. The legacy atlas manuscript and its supplement retain their own artifact identity and are not substituted for this source-to-target protocol.

EC.8. Independent Fixed-Seed Block

The v2 block uses law seeds 95800–95807, with the same public candidate bank, paired keyed target streams, and independent verification procedure. The oracle preflight was recorded before target observations and did not remove a seed. The generated table below is read directly from the frozen result artifact.

All eight fixed seeds were retained after the oracle-only preflight; paired final-reward differences were positive on 2, negative on 2, and tied on 4 tasks.

EC.9. Retained Frontend and Native-Transfer Evidence

The historical component study uses the three fixed families FactorShock, InventorySupplyChain, and QueueResourceControl, with 20 algorithmic seeds per family. Its initial policies were frozen before target observations. The following common-frontend backend table retains the eight supplied-frontend transfer methods. The native table retains each method's own initialization. Both use the historical v69 terminal protocol; the search and source budgets are separate from terminal verification calls. Their totals are not pooled with the current physical-Inventory mean/risk study.

The compact source summaries are `paper_artifacts/paper_result_audit_summary_v1.csv` and `paper_artifacts/or_review/native_transfer.json`. `performance/render_legacy_`

Table EC.7 Historical supplied-frontend backend comparison. Every transfer method receives the frozen frontend, 384 source calls and 13 target search calls.

Method	Feasible	Certified	False	Med. regret	Mean verify
Atlas + canonical SAASBO	60/60	60/60	0	0.0038	173.9
Safe F-PACOH	56/60	56/60	0	0.0082	215.5
RGPE	59/60	59/60	0	0.0082	154.9
Stacked GP	60/60	60/60	0	0.0082	90.9
Transfer MTGP	57/60	57/60	0	0.0082	204.3
FSBO	59/60	59/60	0	0.0082	181.6
HyperBO	60/60	60/60	0	0.0082	211.7
MetaBO	60/60	60/60	0	0.0082	158.9
MALIBO	60/60	60/60	0	0.0082	162.9

Table EC.8 Historical native transfer comparison. Each method retains its own initialization and adaptation with the same source and search budgets.

Native method	FactorShock	Inventory	Queue	Total	False
Safe F-PACOH	0/20	0/20	0/20	0/60	0
RGPE	14/20	16/20	0/20	30/60	0
Stacked GP	0/20	13/20	0/20	13/60	0
MTGP	1/20	0/20	0/20	1/60	0
FSBO	6/20	2/20	0/20	8/60	0
HyperBO	0/20	0/20	0/20	0/60	0
MetaBO	19/20	9/20	0/20	28/60	0
MALIBO	12/20	2/20	0/20	14/60	0

`frontend_evidence.py` generates the aggregate frontend table and its evidence mapping without starting a simulator. The detailed frontend factorial is retained at `tables/frontend_backend.tex`. The method versions are recorded in `performance/manifests/crossdim_frozen_proposal_backend_matrix_v1.json` and `performance/manifests/v9_equal_terminal_verifier_crossdim_gate_v1.json`.

The development integration check in the main text is run by `performance/run_legacy_joint_bridge.py`; its fixed contract is `performance/manifests/legacy_joint_bridge_v1.json`. The compact extracted source observations and seed-80 designs, per-arm paid trajectories, solver status and terminal outcomes are retained under `paper_artifacts/legacy_joint_bridge_v1`. The original 120-iteration run is retained separately from the completed 512-iteration source fit; the latter changes the numerical iteration allowance, with unchanged likelihood and optimization tolerances. All nine source

heads converge. Each complete run uses 48 actual search episodes, including shared initial observations, and 2,096 terminal episodes across the recorded stages. Thus the two runs use 96 search and 4,192 terminal episodes in total, with zero new source calls. Post-decision diagnosis is reproduced by `performance/analyze_legacy_joint_bridge.py`, which makes no new calls. These are development runs on three fixed laws with one seed per law.